Question			Answer	Marks	AOs		Guidance
12	(a)		$0.94^n < k$ or $0.06^n < k$ seen	M1	3.4		allow = instead of <
			$k = 0.025$ used in inequality as above	B1	1.1		
			60	A1	2.2a	NB 59.617.. or 1.311…to 1 or more dp if unsupported implies **M1B1**	60 unsupported or from trial and improvement scores 3
				[3]			

Question			Answer	Marks	AOs		Guidance
12	(b)		$H_0 : p = 0.06$ allow equivalent in words $H_1 : p \neq 0.06$	**B1**	**1.1**		
			p is the probability that a **jaguar** chosen at random is a **black** panther / has **black** coat	**B1**	**2.5**	**or** p is the proportion of **jaguars** that are **black** panthers / have a **black** coat	
			use of B(83, 0.06) to obtain P($X \leq K$) oe	**M1***	**3.3**	not P($X = K$)	NB P($X \leq 10$) = .98927…
			cdfBinomial(83, 0.06, 9) = 0.973 to 0.97321… or 1 – cdfBinomial(83, 0.06, 9) = 0.02679 to 0.027	**A1**	**1.1**	or critical region is $X \geq 11$ (ignore lower tail)	for comparison of their P($X > K$) with 0.025 or their P($X \leq K$) with 0.975 or stating whether 10 is in their critical region
			1 – their P($X \leq K$) compared with 0.025 or their P($X \leq K$) compared with 0.975 oe	**M1dep***	**3.4**	eg 10 compared with their critical region oe	
			result is not significant or do not reject H_0 or reject H_1	**A1**	**1.1**	must have the correct probability or correct critical region for the last two **A** marks	allow accept H_0
			there is insufficient evidence at the 5% level to **suggest** that the probability that a **jaguar** selected at random from this population is a **black panther** is not 0.06	**A1** **[7]**	**2.2b**		do not allow eg conclude / prove / indicate or other assertive statement instead of suggest